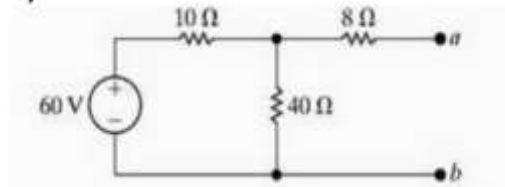


TEOREMAS DE CIRCUITOS – TEOREMAS DE THÉVENIN E DE NORTON – EXEMPLOS (resolução)

Para os circuitos apresentados nas questões 1 a 3, obter os equivalentes de Thévenin e de Norton a partir dos pontos A e B.

1)



SOLUÇÃO

- cálculo de R_{th} :



(fonte anulada)

resistência equivalente vista pelos pontos

A e B:

$$\text{paralelo: } R_p = \frac{10 \times 40}{10 + 40} \Rightarrow R_p = 8\Omega$$

$$\text{série: } R_{th} = 8 + 8 \Rightarrow \boxed{R_{th} = 16\Omega}$$

- cálculo de V_{th} :



(pontos A e B em circuito aberto)

o resistor de 8Ω está em aberto, portanto a tensão sobre ele será zero

a tensão procurada é a tensão sobre o resistor de 40Ω

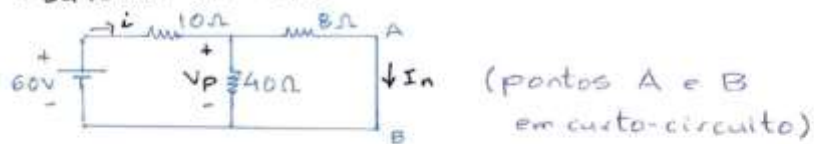
resistência equivalente vista pela fonte:

$$\text{série: } R_{eq} = 10 + 40 \Rightarrow \boxed{R_{eq} = 50\Omega}$$

$$\text{corrente } i: i = \frac{60}{50} \Rightarrow i = 1,2A$$

$$V_{th} = 40i = 40 \times 1,2 \Rightarrow \boxed{V_{th} = 48V}$$

- cálculo de I_n :



resistência equivalente vista pela fonte:

$$\text{paralelo: } R_p = \frac{40 \times 8}{40 + 8} \Rightarrow R_p = \frac{20}{3} \Omega$$

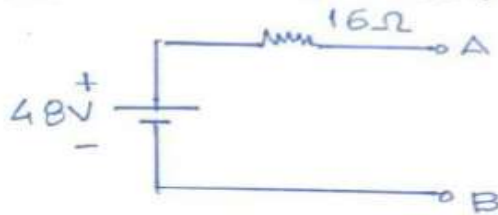
$$\text{série: } R_{eq} = 10 + \frac{20}{3} \Rightarrow R_{eq} = \frac{50}{3} \Omega$$

$$i = \frac{60}{R_{eq}} = 60 \times \frac{3}{50} \Rightarrow i = 3,6 \text{ A}$$

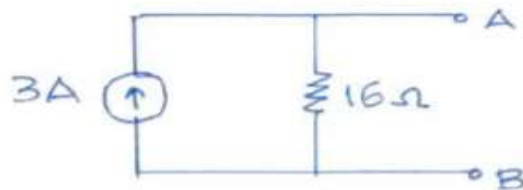
$$V_p = R_p \cdot i = \frac{20}{3} \times 3,6 \Rightarrow V_p = 24 \text{ V}$$

$$\Rightarrow I_n = \frac{V_p}{8} = \frac{24}{8} \Rightarrow I_n = 3 \text{ A}$$

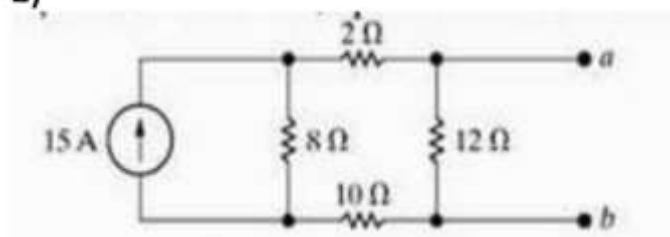
$\Rightarrow$ EQUIVALENTE THÉVENIN:



EQUIVALENTE NORTON:



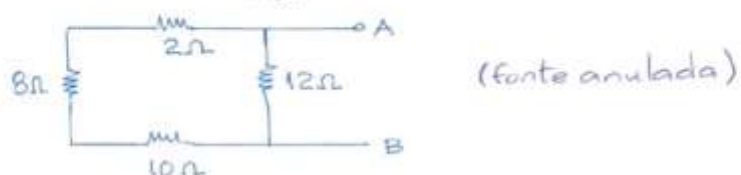
2)



SOLUÇÃO

SOLUÇÃO

- cálculo de R_{th} :



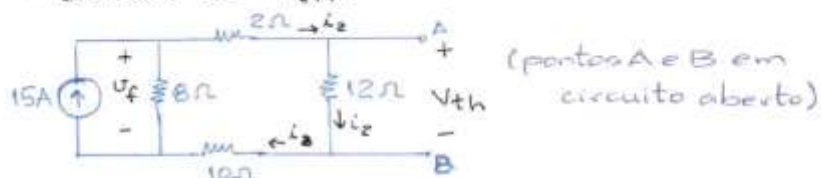
resistência equivalente vista pelos pontos A e B:

$$\text{série: } R_s = 8 + 2 + 10 \Rightarrow R_s = 20\Omega$$

paralelo:

$$R_{th} = \frac{20 \times 12}{20 + 12} \Rightarrow R_{th} = 7,5\Omega$$

- cálculo de V_{th} :



resistência equivalente vista pela fonte:

$$\text{série: } R_s = 2 + 12 + 10 \Rightarrow R_s = 24\Omega$$

$$\text{paralelo: } R_{eq} = \frac{24 \times 8}{24 + 8} \Rightarrow \boxed{R_{eq} = 6\Omega}$$

tensão sobre a fonte de corrente:

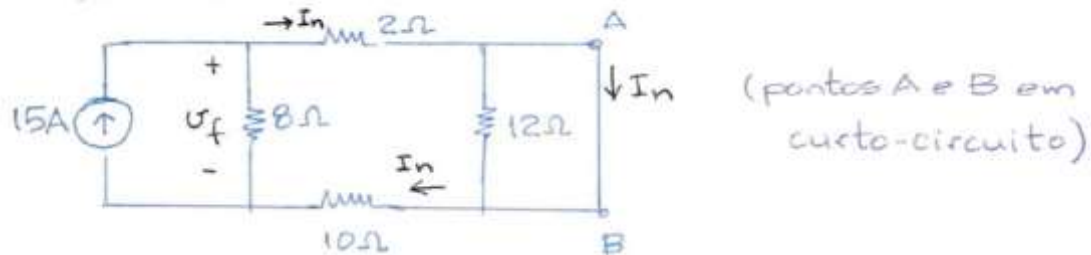
$$U_f = R_{eq} \times 15 = 6 \times 15 \Rightarrow U_f = 90V$$

corrente i_2 :

$$i_2 = \frac{U_f}{R_s} = \frac{90}{24} \Rightarrow i_2 = 3,75A$$

$$\Rightarrow V_{th} = 12 i_2 = 12 \times 3,75 \Rightarrow \boxed{V_{th} = 45V}$$

- cálculo de I_n :



o resistor de 12Ω está em paralelo com o curto-circuito

Portanto, a resistência equivalente vista pela fonte será:

$$\text{série: } R_s = 2 + 10 \Rightarrow R_s = 12\Omega$$

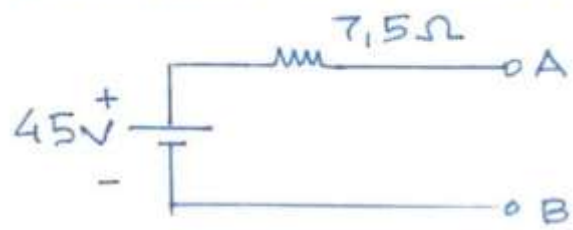
$$\text{paralelo: } R_{eq} = \frac{12 \times 8}{12 + 8} \Rightarrow \boxed{R_{eq} = 4,8\Omega}$$

tensão sobre a fonte de corrente:

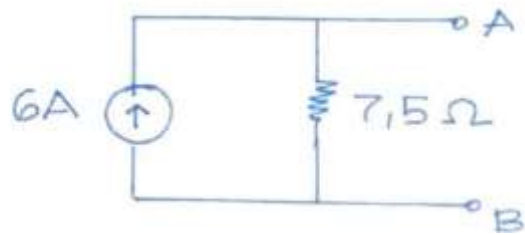
$$U_f = R_{eq} \cdot 15 = 4,8 \times 15 \Rightarrow U_f = 72V$$

$$\Rightarrow I_n = \frac{U_f}{R_s} = \frac{72}{12} \Rightarrow \boxed{I_n = 6A}$$

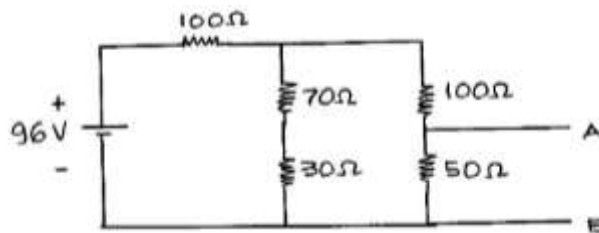
EQUIVALENTE THEVENIN:



EQUIVALENTE NORTON:

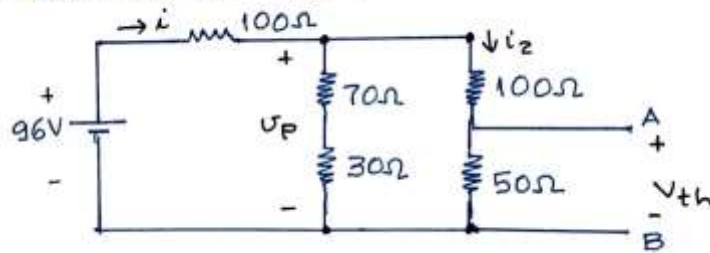


3)



SOLUÇÃO

- cálculo de V_{th} :



série:

$$R_{s1} = 100 + 50 \Rightarrow R_{s1} = 150\Omega$$

$$R_{s2} = 70 + 30 \Rightarrow R_{s2} = 100\Omega$$

paralelo:

$$R_p = \frac{100 \times 150}{100 + 150} \Rightarrow R_p = 60\Omega$$

série:

$$R_{eq} = 100 + 60 \Rightarrow R_{eq} = 160\Omega$$

$$i = \frac{96}{R_{eq}} = \frac{96}{160} \Rightarrow i = 0,6A$$

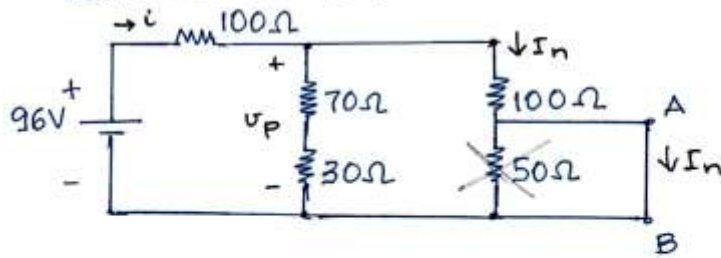
$$V_p = R_p \cdot i = 60 \times 0,6 \Rightarrow V_p = 36V$$

$$i_2 = \frac{V_p}{R_{s1}} = \frac{36}{150} \Rightarrow i_2 = 0,24A$$

$$\Rightarrow V_{th} = 50 i_2 = 50 \times 0,24 \Rightarrow$$

$$V_{th} = 12V$$

- cálculo de I_n :



série: $R_s = 70 + 30 \Rightarrow R_s = 100\Omega$

paralelo:

$$R_p = \frac{100 \times 100}{100 + 100} \Rightarrow R_p = 50\Omega$$

série:

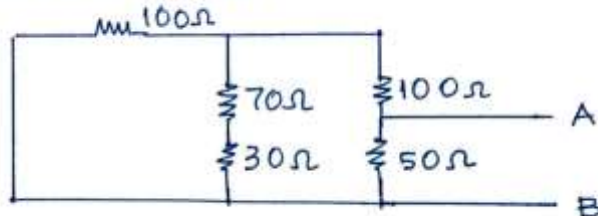
$$R_{eq} = 100 + 50 \Rightarrow \boxed{R_{eq} = 150\Omega}$$

$$i = \frac{96}{R_{eq}} = \frac{96}{150} \Rightarrow i = 0,64\text{A}$$

$$v_p = R_p \cdot i = 50 \times 0,64 \Rightarrow v_p = 32\text{V}$$

$$I_n = \frac{v_p}{100} = \frac{32}{100} \Rightarrow \boxed{I_n = 0,32\text{A}}$$

- cálculo de R_{th} :



série: $R_{s1} = 70 + 30 \Rightarrow R_{s1} = 100\Omega$

paralelo: $R_{p1} = \frac{100 \times 100}{100 + 100} \Rightarrow R_{p1} = 50\Omega$

série: $R_{s2} = 50 + 100 \Rightarrow R_{s2} = 150\Omega$

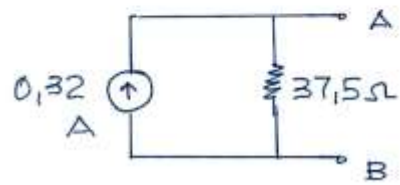
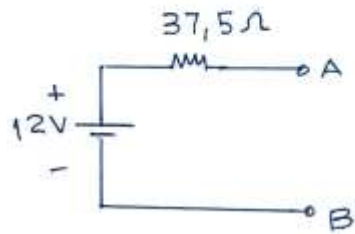
paralelo:

$$R_{th} = \frac{150 \times 50}{150 + 50} \Rightarrow \boxed{R_{th} = 37,5\Omega}$$

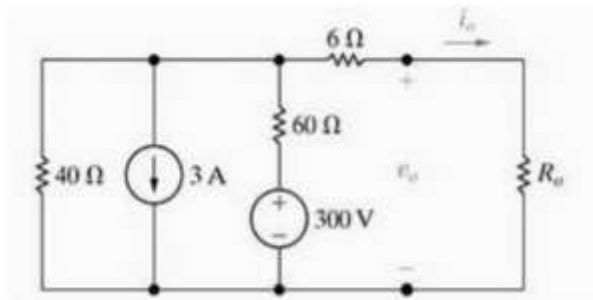
$$\underline{V_{th} = 12\text{ V}}$$

$$\underline{I_n = 0,32\text{ A}}$$

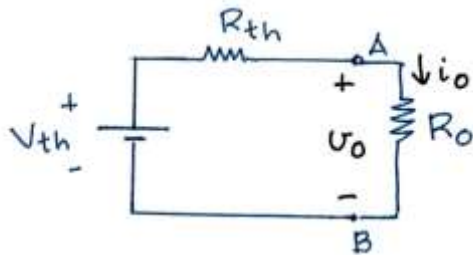
$$\underline{R_{th} = 37,5\ \Omega}$$



4) Para o circuito a seguir, determinar i_0 e v_0 , considerando os seguintes valores para R_0 : $2\ \Omega$, $10\ \Omega$, $50\ \Omega$ e $120\ \Omega$.

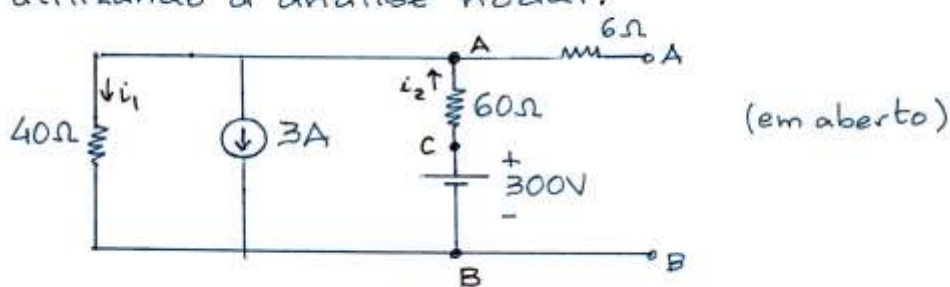


SOLUÇÃO



- cálculo de V_{th} :

utilizando a análise nodal:



referência: nó B

nó C: $v_c = 300V$

nó A:

$$i_1 + 3 = i_2$$

$$\frac{v_A}{40} + 3 = \frac{v_c - v_A}{60}$$

$$\Rightarrow 3v_A + 360 = 2v_c - 2v_A$$

$$\Rightarrow 5v_A = 2v_c - 360$$

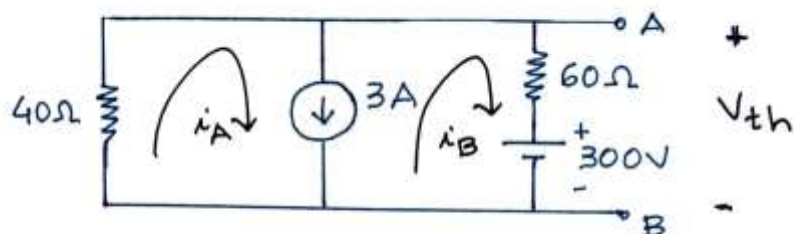
substituindo $v_c = 300$:

$$\Rightarrow 5v_A = 600 - 360 = 240$$

$$\Rightarrow 5v_A = \frac{240}{5} \Rightarrow v_A = 48V$$

$$V_{th} = 5A \Rightarrow \boxed{V_{th} = 48V}$$

utilizando análise de malhas:



fonte de corrente:

$$i_A - i_B = 3$$

percurso externo (supermalha):

$$40i_A + 60i_B + 300 = 0$$

$$\Rightarrow 40i_A + 60i_B = -300$$

$$\begin{cases} i_A - i_B = 3 \xrightarrow{\times 60} 60i_A - 60i_B = 180 \\ 40i_A + 60i_B = -300 \end{cases}$$

$$\Rightarrow 100i_A = -120 \Rightarrow i_A = \frac{-120}{100}$$

$$\Rightarrow \boxed{i_A = -1,2A}$$

$$\Rightarrow i_B = i_A - 3$$

$$\Rightarrow \boxed{i_B = -4,2A}$$

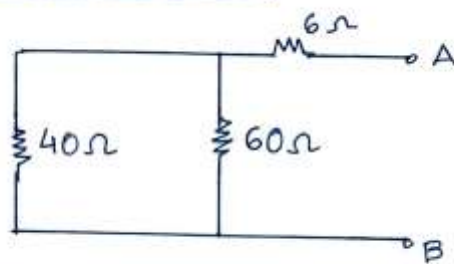
$$V_{th} = 60i_B + 300$$

ou

$$V_{th} = -40i_A$$

$$\Rightarrow \boxed{V_{th} = 48V}$$

- cálculo de R_{th} :



(fontes anuladas)

paralelo:

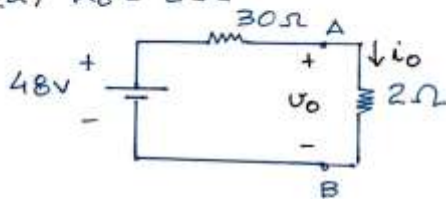
$$R_p = \frac{40 \times 60}{40 + 60} \Rightarrow R_p = 24 \Omega$$

série:

$$R_{th} = 24 + 6 \Rightarrow \boxed{R_{th} = 30 \Omega}$$

- cálculo de v_o e i_o utilizando o equivalente Thévenin

(a) $R_o = 2 \Omega$

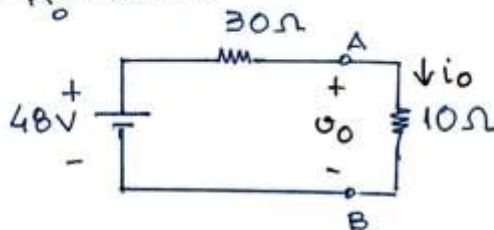


$$\text{série: } R_{eq} = 30 + 2 \Rightarrow R_{eq} = 32 \Omega$$

$$i_o = \frac{48}{R_{eq}} = \frac{48}{32} \Rightarrow \boxed{i_o = 1,5 A}$$

$$v_o = 2i_o = 2 \times 1,5 \Rightarrow \boxed{v_o = 3 V}$$

(b) $R_o = 10 \Omega$

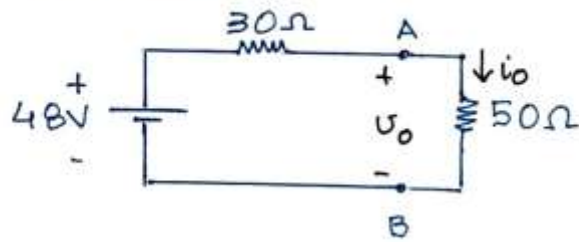


$$\text{série: } R_{eq} = 30 + 10 \Rightarrow R_{eq} = 40 \Omega$$

$$i_o = \frac{48}{R_{eq}} = \frac{48}{40} \Rightarrow \boxed{i_o = 1,2 A}$$

$$v_o = 10i_o = 10 \times 1,2 \Rightarrow \boxed{v_o = 12 V}$$

(c) $R_0 = 50\Omega$

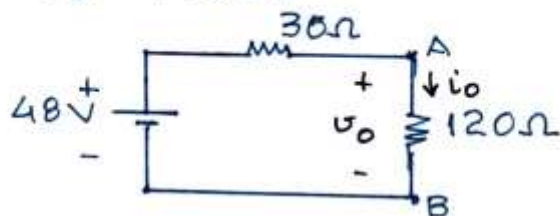


série: $R_{eq} = 30 + 50 \Rightarrow R_{eq} = 80\Omega$

$$i_o = \frac{48}{80} \Rightarrow \boxed{i_o = 0,6\text{A}}$$

$$v_o = 50 i_o = 50 \times 0,6 \Rightarrow \boxed{v_o = 30\text{V}}$$

(d) $R_0 = 120\Omega$



série: $R_{eq} = 30 + 120 \Rightarrow R_{eq} = 150\Omega$

$$i_o = \frac{48}{R_{eq}} = \frac{48}{150} \Rightarrow \boxed{i_o = 0,32\text{A}}$$

$$v_o = 120 i_o = 120 \times 0,32 \Rightarrow \boxed{v_o = 38,4\text{V}}$$